Abstract

Small and medium-sized clothing merchants face significant challenges in digital transformation, most notably the high cost of building independent e-commerce stores, the difficulty of managing products, orders, and inventory, and the absence of unified Arabic platforms that bring them together while maintaining a smooth shopping experience for customers.

This project presents a multi-vendor e-commerce platform for selling clothing, built using Laravel 12, PHP 8.2, and MySQL 8, with interactive front-end interfaces based on Tailwind CSS v4 and Alpine.js. The platform supports four main roles: the administrator who manages users and stores and oversees approvals, the merchant who manages their products, inventory, orders, and coupons, the customer who shops, reviews, and tracks orders, and the visitor who browses without registration.

Key features include browsing products with advanced filters, a dynamic shopping cart, secure checkout, discount coupons, a wishlist, verified reviews conditional on purchase, order status tracking, and an email and in-app notification system. The platform also fully supports the Arabic language with right-to-left layout.

The project followed the Software Development Life Cycle (SDLC) methodology with an Agile model, with complete documentation including the Software Requirements Specification (SRS), ERD diagrams, class and sequence diagrams, and a comprehensive test plan. The system was deployed on the Render cloud platform.

Results demonstrated the system's success in meeting all targeted functional and non-functional requirements, with response times under two seconds, concurrent user support, and a smooth user experience across different devices. The project recommends future expansion to include electronic payment, a mobile application, and a push notification system.

Keywords: E-Commerce, Multi-Vendor, Laravel, Inventory Management, Arabic Platform, Clothing Retail.